

CUSTOMER-READY 8D DEMO

Automotive 8D - CNC Housing Burr at Customer Incoming Inspection

8D-2026-AUTO-001 | Rev.1 | Automotive machining

Approved, unlocked for customer-requested verification evidence, then approved as Rev.1.

Customer / Product / Lot

Northstar Automotive Systems / Aluminum valve housing, part VH-4421 / AH-260418

D1 Team

Maria Chen, Quality Manager; CNC Process Engineer; Production Supervisor; Metrology Technician; Customer Quality Engineer

D2 Problem Description

The customer found a sharp burr on the oil-channel cross hole of 7 CNC-machined aluminum housings from lot AH-260418. The burr exceeded the agreed visual and tactile acceptance standard and could detach during assembly.

D3 Containment

Stopped shipment, quarantined 1,920 pieces across four related lots, introduced 100% borescope inspection, and dispatched certified replacement stock.

D4 Occurrence Cause

The deburring brush reached end-of-life before the scheduled replacement interval because the actual cross-hole load was higher than the interval study assumed.

D4 Escape Cause

Final visual inspection used top lighting only and did not include a borescope angle capable of revealing the cross-hole burr.

D4 Confirmed Root Cause

Inadequate brush-life control caused the burr; an insufficient final inspection angle allowed escape.

D4 5-Why

1. A sharp burr remained because the deburring brush no longer removed the full edge.
2. The brush was used beyond its effective life for this feature.
3. The replacement interval was set from nominal cycles rather than measured burr trend.
4. The control plan had no burr-trend reaction limit.
5. The process risk review did not connect tool wear data to the control-plan reaction plan.

D5 Corrective Action

Set brush replacement at 7,500 cycles, add burr-height reaction limit, and require borescope inspection of the cross hole.

D6 Verification

Three consecutive production lots plus seeded-defect detection challenge.
Lots AH-260503, AH-260506, and AH-260509 passed; zero burr defects in 1,440 parts and all seeded defects detected.

D7 Prevention

PFMEA occurrence and detection controls revised; tool-life reaction rule added.
Applied the same tool-life study to six similar cross-hole machining operations.

D8 Approval

Prepared: Maria Chen (2026-05-10)
Reviewed: David Ruiz (2026-05-11)
Approved: Elena Morris (2026-05-12)

Attachment List

D2: D2-customer-burr-evidence.jpg
D2: D2-customer-defect-photo-description.txt
D4: D4-brush-wear-study.csv
D6: D6-validation-summary.txt

D2 Image Evidence

D2-customer-burr-evidence.jpg

